Let $P_T(x) = (x-\lambda_1)^{m_1}\cdots(x-\lambda_r)^{m_r}$ for operator T on f.dim v.space V over field F

① We saw (by Fitting) $V = \bigoplus_{i=1}^{r} \underbrace{\ker(T-\lambda_i I)^{m_i}}_{V_{\lambda_i}^{gen}}$ ⟹ gen. espace decomp into T-stable subspaces.

② Claim For nilpotent N on any fdvsp W, we have T-stable decomposition $W = \bigoplus_{j=1}^{k} W_j$ such that $N|_{W_j}$ has a staircase matrix  i.e. W_j has a basis $e_1 \cdots e_t$ for which we have $e_t \xrightarrow{N} e_{t-1} \xrightarrow{N} \cdots \xrightarrow{N} e_2 \xrightarrow{N} e_1 \xrightarrow{N} 0$

Pf Later

③ Applying ② to $W = V_{\lambda_i}^{gen}$ with $N = (T-\lambda_i I)|_W$, the macro decomposition in ① breaks further into microdecompositions, giving a basis for V wit which T has block diagonal matrix with each block of Jordan type  .

Claim The multiset of Jordan block sizes in ① (and hence in ③) is unique

Pf. Suppose N has J-form matrix with k_1 blocks of size 1×1 i.e. $\begin{bmatrix}0\end{bmatrix}$
k_2 blocks of size 2×2 i.e. $\begin{bmatrix}0&1\\0&0\end{bmatrix}$
k_3 blocks of size 3×3 i.e. $\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}$

Then we have

$$\ker N \xrightarrow{\dim} k_1+k_2+k_3+\cdots$$
$$\ker N^2 \xrightarrow{\dim} k_1+2(k_2+k_3+\cdots)$$
$$\ker N^3 \xrightarrow{\dim} k_1+2k_2+3(k_3+k_4+\cdots)$$
$$\vdots$$

} can solve easily for $k_1, k_2, k_3 \cdots$

Idea for ②:

Induction on $\dim V$

Assume result on $\mathrm{Im}(N)$

Ex. Finish the proof!

A brave new world: linear algebra with lengths (and angles)

We will work over $\mathbb{R}$ (and later also over $\mathbb{C}$) (modeled on dot product)

Vector spaces will have more structure $\rightsquigarrow$ "inner product spaces"

Similar questions: What is a good basis? How to find one?

What are interesting classes of operators? (orthogonal/unitary, self-adjoint, normal)

Some goals: Gram-Schmidt process

* Spectral theorems for such operators!

Key tool: orthogonal projection/complement

Recall dot product on $\mathbb{R}^n$: $v=\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$ $w=\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}$ $v\cdot w := \sum v_i w_i \in \mathbb{R}$

$= v^t w$ (matrix product)

Def^n ① $\|v\| = \sqrt{v\cdot v}$ length/norm of v

② $\begin{cases} v \perp w \iff v\cdot w = 0 & \text{orthogonality} \\ \cos\theta = \dfrac{v\cdot w}{\|v\|\,\|w\|} \text{ for } v,w\neq 0 & \text{angle between } v \& w \end{cases}$

Ex. This defines a unique $\theta \in [0,\pi]$ b/c $|v\cdot w| \leq \|v\|\|w\|$

Ex Show that this corresponds to actual geometry in $\mathbb{R}^2$ and $\mathbb{R}^3$

Def^n An inner product space/$\mathbb{R}$ is real vector space V with a positive definite symmetric bilinear form $V\times V \to \mathbb{R}$, $v_1, v_2 \to \langle v_1, v_2\rangle$

Properties of dot product

Dot product is ① bilinear ② symmetric ③ positive definite

Pf ① Check $(\alpha\bar{v}+\bar{v}')\cdot\bar{w} = \alpha(\bar{v}\cdot\bar{w})+\bar{v}'\cdot\bar{w}$

similarly in 2nd slot

② $\bar{v}\cdot\bar{w} = \bar{w}\cdot\bar{v}$

③ $\bar{v}\cdot\bar{v} \geq 0$ $\forall \bar{v}$ and $\bar{v}\cdot\bar{v}=0 \Leftrightarrow \bar{v}=\bar{0}$

We call any pos def symmetric bil form on $V/\mathbb{R}$ an inner product on V.

We similarly define $\|\cdot\|$, $\perp$, $\cos\theta$ for such V (after proving Cauchy-Schwarz $\langle v,w\rangle \leq \|v\|\,\|w\|$ Ex)

What is a good basis for an inner product space (say finite dim'l).

Orthonormal basis $v_1,\dots,v_n$ s.t. $v_i\cdot v_j = \delta_{ij}$ i.e. $\|v_i\|=1$ and $v_i \perp v_j$ for $i\neq j$

Example: $e_1,\dots,e_n$ (std basis) for $\mathbb{R}^n$ under dot product.

Note For any $w\in V$ coord of w wrt O.N. basis are easy to get!

$$w = (c_1v_1 + \quad + c_nv_n)\cdot v_i$$

$$w\cdot v_i = \sum_j c_j(v_j\cdot v_i) \implies c_i = w\cdot v_i$$